logging functionality around a method that you want to monitor, you can execute custom code whenever a client inherits from your favorite class…the list goes on. You are limited only by your own, undoubtedly fertile, imagination.

Metaprogramming gives you the power to do all these things. Let's see how this book will help you learn about it.

About This Book

Part I, *Metaprogramming Ruby*, is the core of the book. Chapter 1, *The M Word*, on page 3, walks you through the basic idea behind metaprogramming. The following chapters tell the story of a week in the life of a newly hired Ruby programmer and his or her more experienced colleague:

- Ruby's object model is the land in which metaprogramming lives. Chapter 2, *Monday: The Object Model*, on page 11, provides a map to this land. This chapter introduces you to the most basic metaprogramming techniques. It also reveals the secrets behind Ruby classes and *method lookup*, the process by which Ruby finds and executes methods.
- Once you understand method lookup, you can do some fancy things with methods: you can create methods at runtime, intercept method calls, route calls to another object, or even accept calls to methods that don't exist. All these techniques are explained in Chapter 3, *Tuesday: Methods*, on page 45.
- Methods are members of a larger family also including entities such as blocks and lambdas. Chapter 4, *Wednesday: Blocks*, on page 73, is your field manual for everything related to these entities. It also presents an example of writing a *domain-specific language*, a powerful conceptual tool that Ruby coders tend to love. This chapter also comes with its own share of tricks, explaining how you can package code and execute it later or how you can carry variables across scopes.
- Speaking of scopes, Ruby has a special scope that deserves a close look: the scope of class definitions. Chapter 5, *Thursday: Class Definitions*, on page 105, talks about this scope and introduces you to some of the most powerful weapons in a metaprogrammer's arsenal. It also introduces *singleton classes*, the last concept you need to make sense of Ruby's most perplexing features.
- Finally, Chapter 6, *Friday: Code That Writes Code*, on page 139, puts it all together through an extended example that uses techniques from all the